

Series of exercises No.1

Exercise 1: Understanding the course

a) Convert the following units:

- 2,4 GHz = MHz = Hz
- 4,7 GB = MB = KB = Bytes
- 512 kb/s = KB/s = Bytes/s.
- 2 TB = GB = MB

b) Indicate the method of conversion from **base 10** to **base B**, and vice versa.

c) Indicate the different methods of passing between two bases, which are written in power of 2.

d) Which number system is used in the computer to encode numbers?

e) Indicate on which base(s) the following numbers may belong: **D35F, 879, 1010, 10101I, 4258**

f) In a number system, the biggest digit must equal **(b-1)**, **b** is the base? **(True, False)**

g) Specify the correct formula to find the decimal value of the number $(3B)_{16}$

- ☐ $3 + 11 = (14)_{10}$
- ☐ $3 \times 16 + 1 \times 16 = (64)_{10}$
- ☐ $3 \times 16^1 + 11 \times 16^0 = (59)_{10}$
- ☐ $3 \times 16^1 + 12 \times 16^0 = (60)_{10}$

h) Enter all figures from the base 11

- ☐ {0, 1}
- ☐ {0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B}
- ☐ {0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D}
- ☐ {0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A}

i) Specify the incorrect notations:

- ☐ $(12)_2$
- ☐ $(14)_4$
- ☐ $(187)_{16}$
- ☐ $(AB147)_{16}$

j) How is the number $(9A)_{16}$ written in decimal?

- ☐ 42
- ☐ 22
- ☐ 110
- ☐ None

Exercise 2:

Give the correspondence table of the first 17 integer numbers in the following bases (2, 6, 8, 12, 16)

Exercise 3:

Choose the correct answer:

- 1) $(1830)_{10} =$ **a.** $(2653)_8$ **b.** $(3446)_8$ **c.** $(3448)_8$
- 2) $(1954)_{10} =$ **a.** $(207A)_{16}$ **b.** $(72A)_{16}$ **c.** $(7A2)_{16}$
- 3)** $(2019)_{10} =$ **a.** $(011\ 1101\ 1111)_2$ **b.** $(111\ 1011\ 1110)_2$ **c.** $(111\ 1110\ 0011)_2$

Exercise 4:

Make the following conversions:

- 1) **Base 10 to base X :** **a.** $(69)_{10} = (\dots)_7$ **b.** $(145)_{10} = (\dots)_2$ **c.** $(251)_{10} = (\dots)_{16}$
- 2) **Base X to base 10 :** **a.** $(243)_6 = (\dots)_{10}$ **b.** $(1453)_8 = (\dots)_{10}$ **c.** $(326)_5 = (\dots)_{10}$
- 3) **Base X to Base Y :** **a.** $(6175)_9 = (\dots)_{12}$ **b.** $(234)_5 = (\dots)_7$ **c.** $(1040)_5 = (\dots)_6$

Exercise 5:

Make the following conversions :

- 1) **a.** $(110\ 100)_2 = (\dots)_8$ **b.** $(10\ 011\ 101)_2 = (\dots)_8$ **c.** $(11\ 010\ 100)_2 = (\dots)_8$
- 2) **a.** $(26)_8 = (\dots)_2$ **b.** $(150)_8 = (\dots)_2$ **c.** $(1734)_8 = (\dots)_2$
- 3) **a.** $(1101\ 1000)_2 = (\dots)_{16}$ **b.** $(1001\ 0101\ 1100)_2 = (\dots)_{16}$ **c.** $(1\ 0101\ 0101)_2 = (\dots)_{16}$
- 4) **a.** $(4BF)_{16} = (\dots)_2$ **b.** $(6C2)_{16} = (\dots)_2$ **c.** $(A6E)_{16} = (\dots)_2$
- 5) **a.** $(76)_8 = (\dots)_{16}$ **b.** $(4231)_8 = (\dots)_{16}$
- 6) **a.** $(DCBA)_{16} = (\dots)_8$ **b.** $(D91A)_{16} = (\dots)_8$

Exercise 6:

Determine the base (T, X, Y and Z) in which the following numbers are expressed:

- 1)** $(24)_T = (14)_{10}$ **2)** $(13)_X = (7)_{10}$ **3)** $(70)_Y = (56)_{10}$ **4)** $(1A0)_Z = (416)_{10}$

Exercise 7:

Arithmetic operations on unsigned numbers:

- 1) **a.** $10010 + 111$ **b.** $110110,11 + 1110111,001$ **c.** $10011 - 100$ **d.** $1101101,011 - 101011,11$
- 2) **a.** $(371)_8 + (256)_8$ **b.** $(2EC)_{16} + (7B)_{16}$ **c.** 11110×110 **d.** $1011110 / 101$
- 3)** **a.** $(43)_6 + (34)_6$ **b.** $(AB1)_{16} + (237)_8$

Exercise 8:

- Give the 3 successors of $(626)_8$
- Give the 4 successors of $(E9D)_{16}$
- Give the 5 successors of $(BA9)_{12}$